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Project report - Aerodynamics of Transport Vehicles

Effect of rear wing attachment on Tesla Model S aerodynamic performance

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1. Introduction and Background

The goal of this project is to determine the impact of different rear wing attachment configurations on the aerodynamic performance of a Tesla Model S. The genesis of this work was a visit to the Monza racetrack, where the authors observed a wide range of rear spoiler configurations employed by different racing teams. This observation sparked their curiosity, leading to the following investigation on how the attachment of rear wings can influence a car's downforce, drag, and overall performance.

Aerodynamics plays a crucial role in high-performance vehicle design, particularly in motorsport. Rear spoilers are critical components in achieving optimal aerodynamic balance. Their primary function is to generate *downforce*, which presses the car to the ground, enhancing traction and thus cornering performance. However, this improvement is not free of charge since the addition of a rear wing increases *drag*, reducing top speed, particularly on straights. Therefore, much care is usually taken to achieve maximum downforce and minimum drag, optimizing overall performance.

The following report describes the setup and results obtained with an open-source CFD approach to the problem. CATIA® was chosen for the 3D modeling of the geometry, while OpenFOAM® was used for the meshing and solving steps, and Paraview® was selected for postprocessing.

2. Objectives

Throughout this project, the aerodynamic effect of three different attachment strategies on the same combination of car and rear wing is studied. The entire configuration is studied in unison to properly capture possible interference effects. In particular, the following attachment options are considered:

- **Case 1:** Rear wing attached from below on the wing's suction surface. In what follows, this case is referred to as *Below*.
- **Case 2:** Rear wing attached from above on the aileron's pressure surface, with the structure behind the trailing edge. Throughout this report, this case is denoted as *Back*.
- **Case 3:** Rear wing attached from above on the aileron's pressure surface, with the structure in front of the leading edge. In what remains, the name of this case is *Front*.

Regarding the choice of car, a Tesla Model S is studied. This choice is motivated by the availability of a good quality CAD model for the base car. In addition, it features in different "Gran Turismo" competitions so that this study has some relevance. Regarding the aileron, a 3D model is introduced which combines two airfoils: NACA 4412 in the central section and Selig 1223 on the outboard regions. The former is a classic 4-digit NACA airfoil, while the latter was originally introduced for low Reynolds number flows in UAV's [1], but has also found application in motorsport [2]. The transition between them is done by means of a curved surface, which serves as a vortex generator. Furthermore, endplates are introduced in order to reduce the effect of wingtip vortices and, thus, induced drag.

With respect to the attachments, their geometry was heavily inspired by those readily available online, as well as those observed on the racetrack. All in all, the creation of this geometry and setup of the numerical simulations is described in the following section.

3. Methodology

The aim of this section is to describe the procedure used to analyze the different configurations. First, the 3D modeling process is detailed, providing detailed geometrical information about the different test cases. Then, the setup of the numerical simulations is introduced, focusing on the meshing procedure. Finally, the results of a mesh independence study and the details of the numerical setup are presented.

3.1. Model Creation

The base model for the car was taken from [3] and converted into a CATIA compatible format for editing. There, it was ensured that the geometry was watertight, i.e. with no holes or gaps, so that it could be used for CFD simulations. Then, wheels were added to better represent the car's geometry. Subsequently, different variations of the model were introduced, all with the same wing geometry and placement, but with different attachments. Figures 1 to 5 provide detailed information about the resulting geometry.

Figure 1: 3D CATIA model of Tesla Model S and rear wing

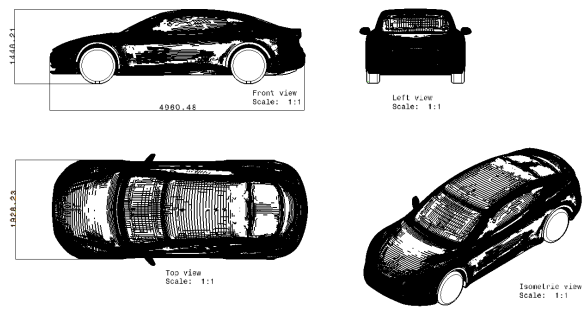

Figure 2: Tesla views and dimensions.

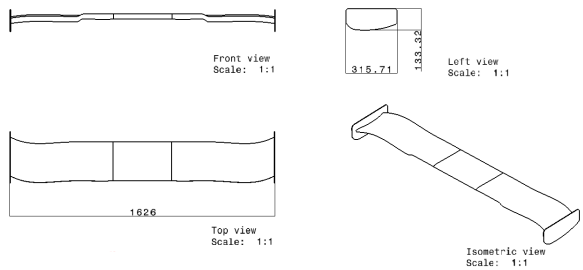

Figure 3: Rear wing views.

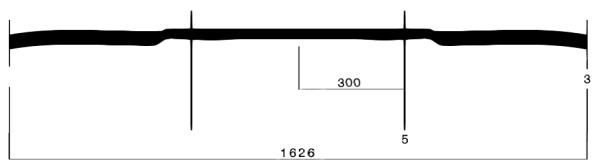

Figure 4: Rear wing frontal view with the attachement.

3.2. Mesh Generation

After the modeling stage, the resulting geometry was exported in *stl* format in order to employ OpenFOAM's mesh generation utilities. The computational domain was initialized using the

(a) Front. (b) Normal. (c) Back.

Figure 5: Different rear wing attachments.

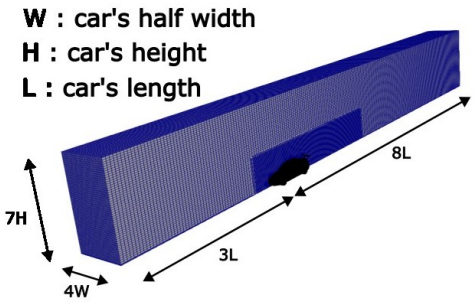

Figure 6: Mesh dimensions.

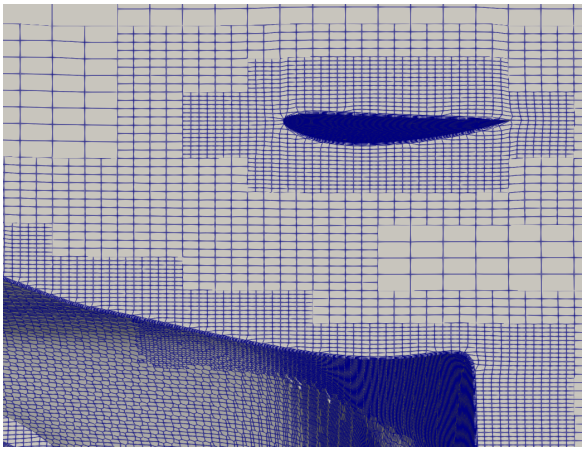

Figure 7: Detail of the mesh on the rear wing.

`blockMesh` utility, which generated a structured, hexahedral background mesh with dimensions sufficient to capture the flow around the vehicle: 3 times the vehicle length (L) upstream, 8 times L downstream, 4 times L laterally, and 7 times the vehicle height (H) above. The mesh was refined using the `snappyHexMesh` utility, resulting in a hybrid mesh. The background mesh remains structured in the far field, while the region near the vehicle transitions to an unstructured, boundary-conforming mesh to capture the complex geometry of the vehicle. This process included surface refinement to resolve sharp edges and critical features, boundary layer mesh generation, and localized refinement in the vicinity of regions where large variations in the flow quantities are expected. The goal of these settings is to balance numerical accuracy and computational cost.

In terms of mesh sizing, different refinement levels were studied in a mesh independence study, the results of which are shown at the end of this section.

3.3. Simulation Setup

Regarding the description of the flow itself, an incompressible RANS approach was chosen to account for the turbulent and separation effects expected at high Reynolds numbers. The resulting governing equations for fluid flow were solved using the `SimpleFoam` solver within the OpenFOAM environment. `SimpleFoam` is a steady-state solver that employs the SIMPLE algorithm to couple pressure and velocity, making it efficient for solving incompressible steady-state flow problems. The `k-omega SST` turbulence model was selected, as it combines the strengths of the `k-omega` model near walls for accurate boundary layer resolution with the robustness of the `k-epsilon` model in the far field.

Thereafter, appropriate boundary conditions were applied to the computational domain:

- **Velocity inlet:** The inflow boundary specified a fixed velocity vector and turbulence properties (`k` and `omega`) to define the incoming flow. The velocity is set to 200 km/h, which is close to the car's maximum speed and thus deemed representative of top speed in a straight. In standard atmospheric conditions at sea level, this results in a Reynolds number of 2.25×10^7, based on the car's length; and 1.1×10^6 based on the rear wing's chord.
- **Pressure outlet:** A static pressure condition was set at the outflow boundary to allow fluid to exit freely with zero gauge pressure.
- **No-slip condition:** Applied to the solid surfaces on the floor and vehicle.
- **Symmetry:** Applied at the plane of symmetry in the real fluid domain. Numerically, as shown in figure 6, only half of the

flow was studied in order to save time and computational resources.

It is important to note that the wheels and ground were modeled as stationary. It is a well-known fact that this makes it impossible to obtain accurate values for aerodynamic forces. Nevertheless, their effect on the flow region around the rear wing is expected to be negligible so that, in relative terms, the conclusions obtained from this study might be applicable, even if the actual values of the forces are not.

3.4. Mesh Independence Study

Based on the previously described setup, a mesh independence study was conducted with the baseline geometry, without a rear wing, to ensure that the simulation results were not influenced by the resolution of the computational grid. Three different levels of mesh refinement were analyzed, whose sizes are presented in table 1. The objective was to verify that aerodynamic parameters, such as lift and drag coefficients, converged to consistent values regardless of further grid refinement.

Table 1: Number of elements (in millions) in the different meshes for the base case without rear spoiler

	Basecase
Coarse	3.2
Medium	5.8
Fine	8.4

Table 2: Relative Errors [%] of C_L and C_D for Mesh Independence Study of baseline configuration at $Re = 2.25 \cdot 10^7$.

	Coarse vs Medium	**Medium vs Fine**
C_L **[%]**	9.17	7.05
C_D **[%]**	3.21	6.95

The results of the mesh independence study show that further refinement of the mesh, beyond the *fine* level, would be necessary to ensure proper mesh convergence of the force coefficients. However, the computational resources necessary for the simulation of the fine mesh already push the limits of the available computational resources. Indeed, 500 iterations with said mesh take 25000 processor seconds, compared with 3475 seconds for the medium level. Furthermore, memory requirements were excessive for some of the available computers. Consequently, it was decided that the medium mesh would be used for the remainder of the project's simulations. As a consequence, the drag and lift values will not agree with true downforce and drag, but this would not happen even for an arbitrarily fine mesh due to the stationary wall and wheels. Therefore, the objective of the following simulations is to highlight the trends and physical differences between different attachments, rather than providing accurate values for their associated lift and drag coefficients.

3.5. Numerical Setup and Validation

The numerical setup of the simulations is heavily inspired by the SimpleFoam motorbike testcase, due to its physical similarities with the problem at hand. Indeed, the turbulence model and initial conditions are taken from those in the test case, as well as the numerical schemes. Convective terms use bounded Gauss Linear Upwind schemes, while diffusive terms are modeled with corrected Gauss linear schemes. In terms of linear solvers, a Generalised-Geometric Algebraic Multigrid solver (GAMG) is used for the pressure equation, while the velocity and turbulence equations are solved with a smooth solver based on Gauss-Seidel smoothing. The numerical behaviour resulting from this setup is presented in figure 8, which shows the evolution of residuals for the baseline configuration using the medium mesh.

Observing figure 8, it is apparent that the residuals stagnate very early on in relatively high values. This stagnation holds for further iterations, effectively stopping at the level shown in the figure. This is a well known issue in the motorbike test case [4]. To understand what is going on, it is interesting to look at the evolution of the force coefficients, represented in figures 9 and 10. In them, it is apparent that once the residuals stagnate the force coefficients exhibit an oscillatory nature around a mean value. Physically, this phenomenology shows that the problem is unsteady, as would be expected of a problem with significant vortex shedding. Thus, the steady

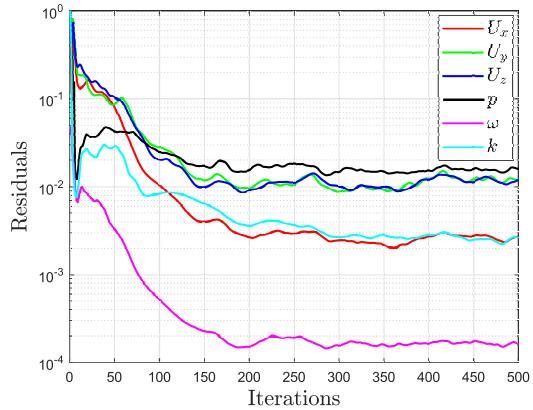

Figure 8: Residuals evolution for Baseline configuration

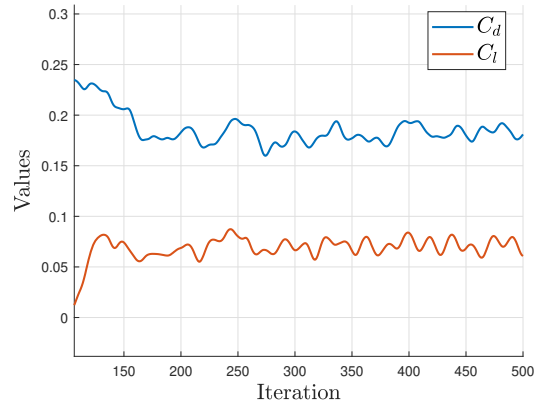

Figure 10: Convergence of C_L and C_D values (Zoom)

solver has difficulty in converging to a steady solution, instead oscillating in a quasi-unsteady state. Consequently, to extract meaningful conclusions from the simulations, it is necessary to look at mean flow values. These values are computed from a 100 iteration mean, taken after the residuals stagnate sufficiently and the force coefficients are well established in their characteristic oscillations. Nonetheless, this should only be seen as a temporary solution and further investigations should focus on a URANS approach to the problem. This approach, while more meaningful, exhibits higher computational cost and was deemed out of the scope of this project.

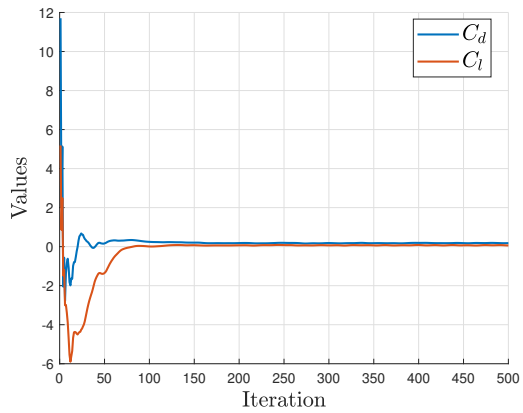

Figure 9: Convergence of C_L and C_D values

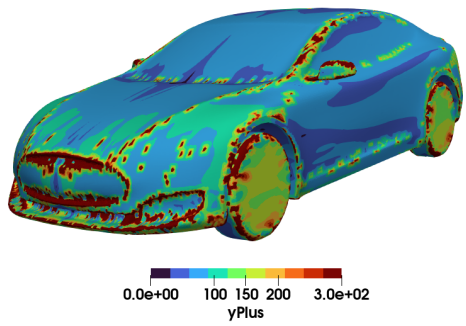

Figure 11: y^+ contour.

On another note, the non-dimensional wall distance y^+ is crucial for assessing the fidelity of turbulence approximations in CFD applications, imposing further restrictions on the mesh. In this study, the k-ω SST turbulence model with wall functions was used, requiring $30 < y^+ < 300$ for accurate results. The results show that, with the aformentioned mesh, low y^+ values occur on the hood, roof, side panels, and upper body, indicating slight over-refinement in some regions. On the other hand, higher values ($y^+ > 200$) appear around the front grill, wheel arches, and lower skirts due to strong pressure gradients and vortex shedding. In conclusion, this shows that the mesh refinement around the vehicle is acceptable. Nevertheless, further refinement of the mesh to balance y^+ along the surface would be an interesting line of inquiry, improving the fidelity of the results as well as the numerical performance of the method.

All in all, this section has presented the workflow followed for the simulations, as well as their limitations and future lines of inquiry. In general, the mean solutions presented in the following section should be interpreted as trend setting comparisons between the different attachments rather than an in-depth assessment of their aerodynamic performance.

4. Results

This section presents the post-processing and analysis of the numerical results obtained from the CFD simulations. The setup used to obtain them, with its limitations, was presented in the previous section. In what follows, an initial comparison between downforce and drag is shown for all cases. Then, a physical explanation for the observed differences is sought by analyzing the flow field in the different configurations.

4.1. Lift and drag coefficients

The purpose of this project is to analyze the effect of the different rear wing attachments on the drag and downforce generated by the car. These are represented by force coefficients based on the car's cross section, estimated at 2.2 m^2. In order to determine this, a baseline case was run without a rear wing and compared with the results obtained with different configurations. The results, shown in table 3 show that the addition of rear wing results in a 200% increase in downforce, justifying its use. This increase comes at a cost of an increase in drag. However, significant differences are seen in the performance of each configuration. To begin with, the *Back* case shows a very slight increase in drag with respect to the base case and a significant increase in downforce, even if it is the lowest of the three rear wing configurations. On the other hand, the *Below* and *Front* configurations lead to very similar results, with downforce values bigger than those of *Back*, paired with a significant increase in drag. Interestingly, however, aerodynamic efficiency is very similar in all cases.

The results shown in table 3 can be interpreted in terms of vehicle dynamics with a very simple model. Assume a curve taken with a load factor n, such that the centrifugal force experienced in it is nW, where W is the weight of the car. Then, this force must be balanced by the grip given by the tires, which depends on the friction coefficient μ and the normal force to the ground. In other words:

Table 3: Comparison of C_L, C_D and Efficiency (E) for each rear spoiler configuration case, for the medium mesh.

	Base	Front	Below	Back
C_L	-0.1120	-0.3506	-0.3513	-0.3268
C_D	0.3640	0.3910	0.3903	0.3642
$\lvert E \rvert$	0.3077	0.8966	0.9000	0.8973

$$nW = \mu(W - L)$$

Where L is the lift provided by the car, computed as $\frac{1}{2}\rho V_c^2 S C_L$. In this formula ρ is the air density, V_c is the car's cornering velocity and S is the reference area taken to compute C_L. Operating on this expression, the value of V_c can be computed as:

$$V_c = \sqrt{\frac{2\left(\frac{n}{\mu} - 1\right)W}{\rho S(-C_L)}}$$

Consequently, the velocity ratio for different configurations in the same turn at the same ambient conditions, approximating the weight as equal, is given by:

$$\frac{V_{c,Modified}}{V_{c,Baseline}} = \sqrt{\frac{C_{L,Baseline}}{C_{L,Modified}}}$$

Thus, a car with more downforce will be able to take a turn with the same load factor at lower velocity or, in turn, take on a higher load factor at the same velocity. Looking at table 4, *Back* shows worse cornering performance than the other configurations, but its lower drag means that it would be a better choice for fast tracks, while *Below* and *Front* are better suited for winding, slower tracks.

Table 4: Cornering velocity ratio for different configurations

	Front	Below	Back
$\frac{V_{c,Modified}}{V_{c,Baseline}}$	0.5652	0.5646	0.5854

Now that the main results in terms of forces have been established, the goal of the investigation is

to provide a physical justification of these results, analyzing the pressure and vortex fields in each case.

4.2. Pressure Results

4.2.1 Pressure distribution at the rear wing

To analyze the aerodynamic behavior of the different rear wing configurations, the mean pressure coefficient, $C_{p,mean}$, was evaluated on both the extrados and intrados of the wing in Fig. 13 and Fig. 14. Two longitudinal cuts were taken at distances of 0.25 m (section A-A) and 0.75 m (section B-B) from the symmetry plane of the car to assess variations along the span, as can be seen in Fig.12.

Figure 12: Sections taken from the rear wing to represent the evolution of $C_{p,mean}$

In both plots the extrados (solid lines) exhibits a characteristic negative pressure peak near the leading edge due to flow acceleration, followed by a gradual recovery towards the trailing edge. The intrados (dashed lines) maintains higher pressure levels, with minimal variations along the chord.

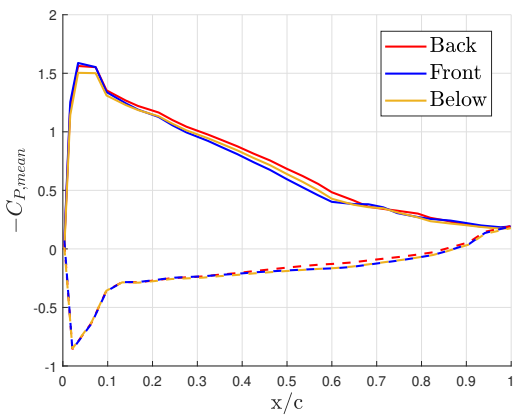

Figure 13: Distribution of $C_{p,mean}$ at rear wing extrados (solid line) and intrados (dashed line), at section A-A

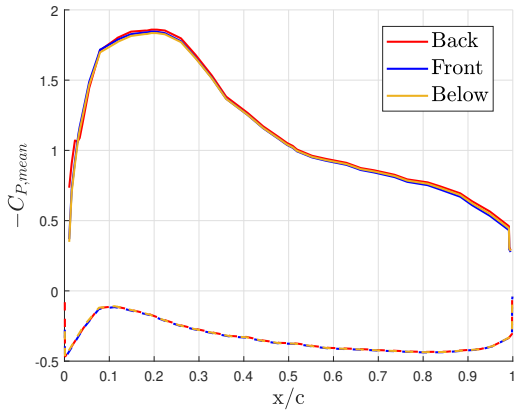

Figure 14: Distribution of $C_{p,mean}$ at rear wing extrados (solid line) and intrados (dashed line), at section B-B

Section A-A shows lower and narrower suction peak in comparison with B-B, which highlights the three dimensional nature of the flow. In classical aerodynamic theory, the opposite would be expected, as the section load distribution becomes less intense as the tips are approached. However, the influx of "dirty" air from the car, the spanwise curvature of the rear wing and the presence of endplates intensify three dimensional effects and lead to a very complex pressure distribution. In fact, relevant interference effects will be a constant throughout the remainder of this work.

Comparing different configurations, the pressure distribution on the outboard region of the wing is unaffected by the different attachments. In contrast, the inboard region shows clear differences. On the pressure surface, the *Back* configuration exhibits lower pressure values than the other two, which show very similar distributions. On the suction surface, the different configurations alternate along the chord, showing a generally complex evolution. All in all, this highlights the complex, three dimensional flow on the rear wing, where the differences in downforce cannot be reduced to a single, simple contribution. As the next section will show, this is not the same for the drag generation on the car's body, where clear justifications for the differences can be found.

4.2.2 Pressure distribution at the rear of the car

The first step in assessing the differences in pressure distribution on the rear of the car is an analysis of its wake. This is done on the symmetry plane by studying the evolution of the total pressure coefficient, defined as the difference between actual and reference total pressure, non-dimensionalised by reference dynamic pressure. Under the conditions of ideal flow, total pressure should be conserved, leading to a null coefficient. On the contrary, on regions with important viscous effects, total pressure should drop, so that a negative total pressure coefficient clearly indicates the existence of the wake. Figure 15 shows this evolution.

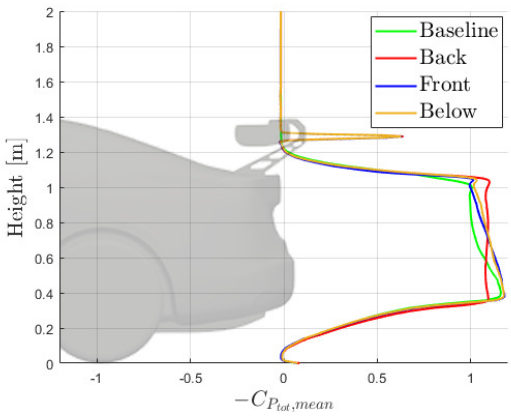

Figure 15: Distribution of $C_{p_{Total},mean}$ right at the rear of the car in the car's plane of symmetry

Focusing on figure 15, two significant wake regions can be identified. First, a clear wake is seen behind the rear wing, whose characteristics are identical in all configurations with such a wing. Secondly, a big wake region is seen at the back of the car, where there is significant flow separation. The intensity of the wake, represented by the absolute value of the total pressure coefficient is actually affected by the different attachments, providing clear evidence of interference effects. The *Back* configuration leads to a more balanced wake, while *Front* and *Below* lead to more intense wakes at the bottom and less intense ones at the top. In all cases, the wake is more intense than that created in the baseline configuration. Finally, the boundary layer at the ground also results in a total pressure reduction in its vicinity, independent of the given configuration.

Inspired by the results shown in the wake analysis, the pressure distribution at the rear of the car is studied, focusing on a cut at a distance of 0.5 m from the ground. Figure 16 shows the evolution of mean pressure coefficient just aft of the car in this plane. Comparing the different configurations, they show different evolutions that illustrate differences in their wake structures. Nonetheless, close to the car, the *Baseline*, *Front*, and *Below* configurations show similar pressure values while *Back* exhibits a significantly higher base pressure and, thus, less pressure drag than the other configurations.

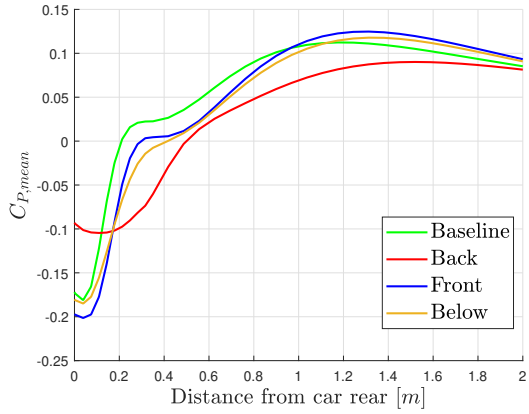

Figure 16: Evolution of $C_{p,mean}$ just after the rear of the car.

In light of the previous results, figure 17 shows the transverse evolution of the mean pressure coefficient on the same plane. The y coordinate is 0 at the symmetry plane, with negative values representing the car's right and positive values its left. All configurations show a clear decrease in pressure on the car's wake, with a gradual recuperation towards the ends of the fluid domain. Focusing on differences in this wake, it is apparent that *Back* exhibits a more stable mean pressure level, whose average on the wake is higher than all other configurations. Furthermore, close to the plane of symmetry, it grows significantly, leading to a relatively high pressure bubble behind the car. This justifies, once more, a lower value of pressure drag. In stark contrast, the *Front* and *Below* configurations indicate a clearly lower mean pressure level when compared with the *Baseline* and *Back* configurations, justifying higher pressure drag.

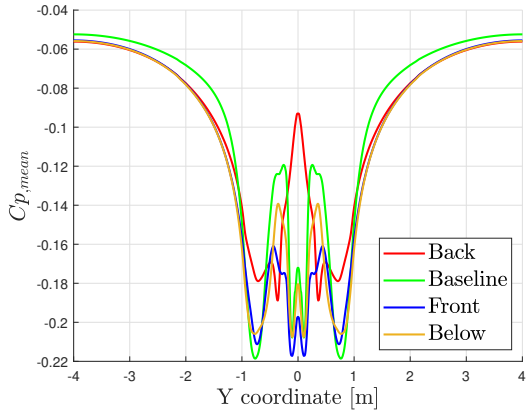

Figure 17: Distribution of $C_{p,mean}$ in transverse direction just after the rear of the car

In conclusion, a pressure analysis on the wake has proven that the different attachments lead to different wake structures, with *Back* reducing pressure drag with respect to *Baseline*, while *Front* and *Below* increase it. Taking this into account, the next section analyses the vortical structures formed in each case in order to further illustrate the physical effect of the different attachments.

4.3. Vortex structures: Q criterion

The visualization of vortical structures in the flow can be very useful in justifying its behavior and understanding the factors that influence the force coefficients obtained in table 3. To this end, contour plots of the Q criterion [5] for vortex identification are employed and shown in figures 18 and 19. In both cases a contour value of 2000 proved to be best at showing the flow structure without excessive clutter.

Assessing figures 18 and 19, some common structures can be identified. Regarding the rear wing, longitudinal vortices are seen at the end plates, representing wingtip vortices; and at the spanwise curved region where the airfoil is changed. In addition, smaller vortices are shown at the trailing edge. These all represent the classic vortex sheet wake structure of a lifting surface. The attachments themselves also lead to some small vortices, but their rather compact shape means that these are not too relevant. With respect to the *Baseline* case, a longitudinal vortex can be seen which extends from the A-pillar through the top of the car, down the C-pillar and finally over the back.

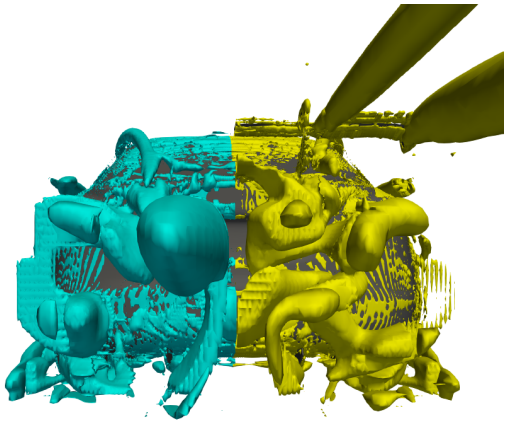

Figure 18: Q criterion: Rear of the car of the *Basecase* (on the left, in blue) and *Back* rear wing configuration (on the right, in yellow)

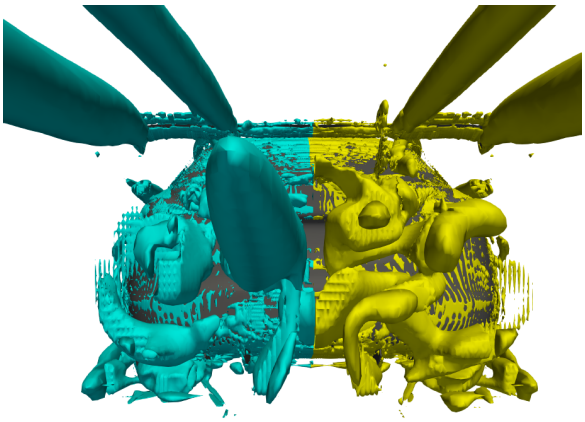

Figure 19: Q criterion: Rear of the car with *Below* rear wing configuration (on the left, in blue) and *Back* rear wing configuration (on the right, in yellow)

On another note, significant differences can be seen in the vortical structures at the rear of the car, particularly between *Back* and the other cases, which are more similar between themselves. In essence, *Baseline*, *Below* and *Front* exhibit a flow with predominant longitudinal vortical structures, whereas for *Back*, significant transverse vorticity is also seen. This means that the separation region behind the *Back* configuration is more akin to a separation bubble of slow moving, higher pressure air, while that seen in other cases is closer to a vortex sheet of air leaving the car, with lower base pressure. This explains the differences in base pressure seen in the previous section and lend creedence to the hypothesis that the difference in drag between *Back* and the other configurations is due to its

lower pressure drag.

4.4. Final analysis of rear wing configurations

Based on the heretofore shown evidence, it is possible to extract some conclusions about the behavior of the flow in each case, as well as its effect on the resulting forces. Firstly, it is apparent that the *Back* configuration leads to a modified wake at the rear of the car, increasing its base pressure and thus reducing pressure drag with respect to the *Baseline* configuration. This reduction is then offset by the extra friction and induced drag resulting from the addition of the rear wing, leading to a drag coefficient similar to that seen in the *Baseline* configuration. Consequently, interference effects are positive in this case. On the other hand, *Front* and *Below* lead to a predominantly longitudinal wake structure similar to that in the *Baseline* case and, thus, do not offer a reduction but an increase in pressure drag. They still reflect in an increase in friction and induced drag, so that, in overall terms, they provide higher drag than *Baseline*. Nevertheless, the flow field generated around the rear wing in the latter two cases leads to higher downforce compared with *Back*, which shows that there is an aerodynamic tradeoff between these configurations.

5. Conclusions

Throughout this report, a numerical approach for the study of rear wing attachments on a race car has been introduced, as well as the motivation behind it. In total, three different attachment geometries were tested for the same combination of car and rear wing, focusing on the resulting downforce and drag coefficients. These results were then justified by a careful analysis of the flow field.

All in all, CFD seems to point at the fact that *Back* is optimal when achieving downforce with minimum drag is the priority, while *Below* and *Front* show similar behavior and trade an increase in downforce with an increase in drag. This indicates that each configuration could have its place in a race depending on the team's strategy and characteristics on the track. However, it is important to bear in mind that these attachments are subject to many requirements beyond those studied in this case. Aerodynamically, they must also fare well in the presence or crosswind and sideslip in general, which was not studied in this case due to time and resource limitations. Moreover, structural, weight and aeroelastic considerations must also be accounted for in order to produce an attachment that best fits the need of the vehicle.

It must also be noted that the numerical results shown in this report should be taken with a grain of salt. As previously discussed, a true evaluation of the aerodynamic performance of the car would require more refined meshes, unsteady simulations and moving ground and wheels. Even in this case, the strong dependence on interference effects seen in this project might imply that a CFD approach is unsuitable in general and wind tunnel tests are the best approach to obtain precise measurement. In conclusion, it must be noted that the results of this project, while interesting with regards to future study, need not hold when the fidelity of the model is increased.

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